Exact Consistency in Multi-Location Encodings: Zero-Error Thresholds and Side Information Bounds

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***Abstract*—We study exact consistency in finite multi-location encodings, where one latent value is represented at several writable or derived locations and an observer sees a deterministic transcript of the current state. Our main contribution is a deterministic finite converse: if the observation transcript leaves a k-way ambiguity class, any zero-error resolver requires at least $\log_2 k$ bits of side information, and the same finite obstruction admits confusability, conditional-entropy, decoder-output, and entropy-gap formulations. As a threshold corollary, the largest independent rate compatible with zero incoherence is exactly 1: one authoritative source with deterministic views achieves exact consistency, whereas more than one independently writable source makes ambiguity reachable. This threshold has an operational consequence for updates, giving $O(1)$ manual modification cost at rate 1 and $\Omega(n)$ cost above it. A supplementary Lean 4 artifact machine-checks the core threshold, converse, realizability, and rate-complexity arguments. Index Terms: zero-error information theory, side information, confusability, exact consistency, deterministic converse**

I. Introduction

When one latent fact is represented at several modifiable locations, exact consistency becomes a zero-error inference problem. The current system state either identifies one authoritative value or leaves a nontrivial ambiguity class of mutually incompatible latent states.

An abstract multi-location encoding model is used throughout. A fact F is represented across locations $\{L_1, \dots, L_n\}$, some independently writable and some derived from others. Two questions drive the analysis:

1) When does the encoding architecture itself guarantee exact consistency?
2) When it does not, how much side information is needed for zero-error resolution?

The resulting theory belongs to the intersection of zero-error information theory, side-information source coding, and finite converses [1]–[6].

A. Problem Formulation

The model yields an exact-consistency analogue of zero-error resolvability for modifiable encodings. Rate is measured by the number of independently writable locations; we refer to this quantity as the *independent rate* and use *degrees of freedom* (DOF) only as shorthand. The fundamental threshold is

$$\boxed{C_0 = 1}$$

where C_0 denotes the largest independent rate that guarantees zero incoherence.

A finite converse family shows that once ambiguity survives the observation transcript, the same finite information budget reappears in several classical forms.

Theorem I.1 (Unified finite converse, informal)**.** *Let X be a finite latent source, let Y be the deterministic observation transcript available to a resolver, and let $\hat{X}$ be any deterministic decoder output. If Y does not isolate the true state, then the same finite observation/side-information budget simultaneously governs exact zero-error resolution of ambiguity classes, bounds the conditional entropy $H(X \mid Y)$, bounds the decoder-output entropy $H(\hat{X})$, and appears as an output-entropy or KL gap relative to the corresponding finite alphabet ceiling. In particular, exact k-way resolution requires at least $\log_2 k$ bits of side information.*

The threshold theorem and the converse family interact: $C_0 = 1$ identifies the unique zero-incoherence regime, while the converse explains what fails once the architecture moves above that regime.

B. Relation to Classical Information Theory

Three classical lines are particularly relevant.

Zero-error information theory. Shannon's zero-error framework and its graph-theoretic refinements describe exact communication under confusability constraints [1]–[3]. Our model translates that perspective from one-shot channels to persistent encodings under edits.

Source coding with side information. In Slepian–Wolf theory, and in Witsenhausen's zero-error side-information problem, decoder side information lowers the rate or label budget needed for correct recovery [4], [5]. Coding-for-computing and functional-compression results make the same structure explicit when the decoder seeks a deterministic function rather than the full source [7], [8]. Here the side information is deterministic rather than stochastic, but it plays the same operational role: exact recovery is impossible unless the available tags or observations separate all remaining alternatives.

Finite entropy converses. Classical Fano inequalities and related entropy bounds convert decoding difficulty into information bounds [6], [9], [10]. Our setting yields a deterministic finite-source analogue in which counting, conditional entropy, decoder-output entropy, and output-gap formulations become different normal forms of the same obstruction.

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C. Realizability and Applications

Beyond the abstract threshold, a second question is when a concrete host system can realize the rate-1 regime. We show that two structural properties are required:

1) **Causal update propagation:** derived locations must update automatically when the source changes.
2) **Provenance observability:** the system must expose enough structural information to verify which locations are authoritative and which are derived.

D. Paper Organization

Section II formalizes the encoding model, the zero-incoherence threshold, and the base side-information lower bound. Section III interprets derivation as the dependence structure achieving rate 1. Section IV gives the realizability theorem. Section VI develops the finite converse family and its rate-complexity consequence. Sections V and VII provide brief computational illustrations. A supplementary Lean 4 artifact machine-checks the core counting, threshold, realizability, and complexity chains [11].

E. Scope

The scope is finite facts represented at multiple locations under admissible edits. The focus is on *structural facts*, meaning facts whose encoding locations are fixed when an object, declaration, or artifact is created. This isolates the zero-error regime most clearly and captures a wide class of realizations without making the theory domain-specific.

F. Contributions

The main contributions are:

- **A deterministic finite converse family** for exact consistency, starting from the $\log_2 k$ side-information lower bound and extending to confusability, conditional-entropy, decoder-output, and entropy-gap formulations.
- **A sharp threshold corollary** showing that zero-incoherence capacity is exactly one independent source.
- **A realizability theorem** identifying causal propagation and provenance observability as the conditions needed to attain the rate-1 regime in concrete systems.
- **An operational consequence** showing $O(1)$ manual update cost at rate 1 and $\Omega(n)$ cost above it, with an unbounded asymptotic gap.

II. Encoding Systems and Coherence

We formalize exact consistency for multi-location encodings with modification constraints. Facts take values in finite alphabets, system states expose multiple representations of those facts, edits move the system among reachable states, and the observer must determine whether the represented fact is uniquely recoverable.

A. Model assumptions and notation

We state the modeling assumptions used throughout the paper.

1) **Fact value model (A1):** Each fact F takes values in a finite set $\mathcal{V}_F$; $H(F) = \log_2 |\mathcal{V}_F|$ denotes its entropy under a chosen prior. When we write "zero probability of incoherence" we mean $\Pr(\text{incoherent}) = 0$ under the model for edits and randomness specified below.
2) **Update model (A2):** Modifications occur as discrete events. An edit δ_F changes the source value for F; derived locations update deterministically when causal propagation is present. We do not require a blocklength parameter; asymptotic statements are in the number of encoding locations n.
3) **Adversary / randomness (A3):** When randomness is needed we assume a benign stochastic edit model for illustration. For impossibility and converse statements we make no cooperative assumptions: adversarial edit sequences that produce incoherence are allowed whenever the independent rate exceeds 1.
4) **Side-information channel (A4):** Derived locations and runtime queries collectively form side information S available to any verifier. In the finite zero-error layer, the relevant quantity is the number of distinguishable tags that S can expose.

References to these assumptions use labels (A1)–(A4). Full formal versions of the operational model are given in the mechanized supplement (Supplement A).

B. The Encoding Model

We begin with the abstract encoding model: locations, values, reachable states, and ambiguity.

Definition II.1 (Encoding System)**.** An *encoding system* for a fact F is a collection of locations $\{L_1, \ldots, L_n\}$, each capable of holding a value for F.

Definition II.2 (Coherence)**.** An encoding system is *coherent* iff all locations hold the same value:

$$\forall i, j : \text{value}(L_i) = \text{value}(L_j)$$

Definition II.3 (Incoherence)**.** An encoding system is *incoherent* iff some locations disagree:

$$\exists i, j : \text{value}(L_i) \neq \text{value}(L_j)$$

Definition II.4 (Ambiguity Class)**.** For a system state x, the *ambiguity class* of x is the set

$$\mathcal{A}(x) = \{v : \exists L \text{ with value}(L) = v\}.$$

When $|\mathcal{A}(x)| > 1$, the state presents multiple internally supported candidate values for the same fact.

The zero-error resolution problem. Given a state x, a resolver must output the correct value of F with zero error. When $|\mathcal{A}(x)| > 1$, the internal state alone may fail to identify a unique answer.

Theorem II.5 (Oracle Arbitrariness)**.** *For any incoherent encoding system state x and any oracle O that resolves x*

to a value $v \in \mathcal{A}(x)$, there exists a value $v' \in \mathcal{A}(x)$ such that $v' \neq v$.

Proof. By incoherence, $|\mathcal{A}(x)| > 1$, so there exist $v_1, v_2 \in \mathcal{A}(x)$ with $v_1 \neq v_2$. Either O picks v_1 (then v_2 disagrees) or it does not pick v_1 (then v_1 disagrees). ■

Definition II.6 (Degrees of Freedom)**.** The *degrees of freedom* (DOF), or *independent rate*, of an encoding system is the number of locations that can be modified independently.

Theorem II.7 (Unit Independent Rate Guarantees Coherence)**.** *If the independent rate is* 1*, then the encoding system is coherent in all reachable states.*

(L: COH1)

Proof. At independent rate 1, exactly one location is independent. All other locations are derived (automatically updated when the source changes). Derived locations cannot diverge from their source. Therefore, all locations hold the value determined by the single independent source. Disagreement is impossible. ■

Theorem II.8 (Independent Rate Above One Permits Incoherence)**.** *If the independent rate exceeds* 1*, then incoherent states are reachable.*

(L: COH2)

Proof. With DOF > 1, at least two locations are independent. Independent locations can be modified separately. A sequence of edits can set $L_1 = v_1$ and $L_2 = v_2$ where $v_1 \neq v_2$. This is an incoherent state. ■

Corollary II.9 (Exact Consistency Forces Unit Rate)**.** *If coherence must be guaranteed, then the independent rate* 1 *is necessary and sufficient.*

This is the basic exact-consistency criterion in the model.

The following subsections refine the model and introduce the quantities used in the later converse statements.

C. Host Systems

The abstract model applies to any host system that exposes a finite family of addressable locations, an admissible edit set, and a correctness condition that can be violated by inconsistent updates.

Definition II.10 (Host System)**.** A *host system* C is a finite collection of addressable locations together with a set $E(C)$ of admissible edits. Each edit $\delta \in E(C)$ transforms the current state of C into another reachable state.

Definition II.11 (Location)**.** A *location* $L \in C$ is an addressable component of the host system whose value may participate in the representation of a fact.

Definition II.12 (Modification Space)**.** For host system C, the *modification space* $E(C)$ is the set of admissible edits on C.

The modification space can be large, but only edits that change one designated fact are considered here.

D. Facts: Atomic Units of Specification

Definition II.13 (Fact)**.** A *fact* F is an atomic semantic attribute of the represented object that can change independently of other attributes.

The granularity of facts is determined by correctness, not by storage layout. If two pieces of information must change together, they form one fact; if they can change independently, they are distinct facts.

Definition II.14 (Structural Fact)**.** A fact F is *structural* with respect to encoding system C iff the locations encoding F are fixed at definition time:

$$\begin{aligned}\text{structural}(F, C) \iff \{L : \text{encodes}(L, F)\} \\ \subseteq \text{DefinitionSyntax}(C)\end{aligned}$$

where $\text{DefinitionSyntax}(C)$ denotes the part of the host-system description that fixes the encoding locations when the relevant object is created.

Structural facts are fixed when the relevant object is created or declared. Once fixed, the structure cannot be repaired retroactively without a new structural edit. Non-structural facts may require reactive or feedback-based mechanisms not studied here.

E. Encoding: The Correctness Relationship

Definition II.15 (Encodes)**.** Location L *encodes* fact F, written $\text{encodes}(L, F)$, iff correctness requires updating L when F changes.

Formally:

$$\begin{aligned}\text{encodes}(L, F) \iff \forall \delta_F :\ \neg\text{updated}(L, \delta_F) \\ \to \text{incorrect}(\delta_F(C))\end{aligned}$$

where δ_F is an edit targeting fact F.

F. Modification Complexity

Definition II.16 (Modification Complexity)**.**

$$M(C, \delta_F) = |\{L \in C : \text{encodes}(L, F)\}|$$

The number of locations that must be updated when fact F changes.

Modification complexity is the central operational metric of the paper. It measures the cost of changing a fact in terms of locations that must be updated for correctness.

Theorem II.17 (Correctness Forcing)**.** *$M(C, \delta_F)$ is the* ***minimum*** *number of edits required for correctness. Fewer edits imply an incorrect system state.*

Proof. Suppose $M(C, \delta_F) = k$, meaning k locations encode F. By Definition II.15, each encoding location must be updated when F changes. If only $j < k$ locations are updated, then $k - j$ locations still reflect the old value of F. These locations create inconsistencies:

1) The specification says F has value v' (new)
2) Locations $L_1, \ldots, L_j$ reflect v'
3) Locations $L_{j+1}, \ldots, L_k$ reflect v (old)

By Definition II.15, the resulting state is incorrect. Therefore, all k locations must be updated, and k is the minimum. ■

G. Independence and Degrees of Freedom

Not all encoding locations are created equal. Some are *derived* from others.

Definition II.18 (Independent Locations)**.** Locations L_1, L_2 are *independent* for fact F iff they can diverge. Updating L_1 does not automatically update L_2, and vice versa.

Formally: L_1 and L_2 are independent iff there exists a sequence of edits that makes L_1 and L_2 encode different values for F.

Definition II.19 (Derived Location)**.** Location L_{derived} is *derived from* L_{source} iff updating L_{source} automatically updates L_{derived}. Derived locations are not independent of their sources.

Definition II.20 (Degrees of Freedom)**.**

$$\text{DOF}(C, F) = |\{L \in C : \text{encodes}(L, F) \wedge \text{independent}(L)\}|$$

The number of *independent* locations encoding fact F, i.e., the independent rate for that fact.

The independent rate is the key metric. Modification complexity M counts all encoding locations. DOF counts only the independent ones. If all but one encoding location is derived, the independent rate remains 1 even though M can be large.

Theorem II.21 (DOF = Incoherence Potential)**.** *DOF$(C, F) = k$ implies that up to k distinct values for F can coexist simultaneously. In particular, $k > 1$ makes ambiguity reachable.*

Proof. Each independent location can hold a different value. By Definition II.18, no constraint forces agreement between independent locations. Therefore, k independent locations can simultaneously realize up to k distinct values. ■

Corollary II.22 (DOF > 1 Implies Incoherence Risk)**.** *DOF$(C, F) > 1$ implies incoherent states are reachable.*

H. Information-Theoretic Quantities

Before establishing the main threshold result, we define the information-theoretic quantities used to express ambiguity, redundancy, and side-information requirements.

Definition II.23 (Independent Rate)**.** The *independent rate* of system C for fact F is $R(C, F) = \text{DOF}(C, F)$, the number of independent encoding locations.

Definition II.24 (Value Entropy)**.** For a fact F with value space $\mathcal{V}_F$, the *value entropy* is:

$$H(F) = \log_2 |\mathcal{V}_F|$$

This is the information content of specifying F's value.

Definition II.25 (Encoding Redundancy)**.** The *encoding redundancy* of system C for fact F is:

$$\rho(C, F) = \text{DOF}(C, F) - 1$$

Redundancy measures excess independent encodings beyond the minimum needed to represent F.

Theorem II.26 (Redundancy-Incoherence Equivalence)**.** *Encoding redundancy $\rho > 0$ is necessary and sufficient for incoherence reachability:*

$$\rho(C, F) > 0 \iff \textit{incoherent states reachable}$$

Proof. ($\Rightarrow$) If $\rho > 0$, then DOF > 1. By Theorem II.8, incoherent states are reachable.

($\Leftarrow$) If incoherent states are reachable, then by contrapositive of Theorem II.7, DOF $\neq 1$. Since the fact is encoded, DOF ≥ 1, so DOF > 1, hence $\rho > 0$. ■

Definition II.27 (Incoherence Entropy)**.** For a state whose ambiguity class has size k, the *incoherence entropy* is:

$$H_{\text{inc}} = \log_2 k$$

This quantifies the uncertainty about which value is authoritative.

I. Zero-Incoherence Capacity Theorem

We now establish the central exact-consistency threshold. The argument is combinatorial, and its role is to expose the single-source regime on which the later converse and rate-complexity results are built.

Background: Zero-Error Capacity. Shannon [1] introduced zero-error capacity: the maximum rate at which information can be transmitted with *zero* probability of error. Körner [2] and Lovász [3] linked such questions to confusability structure. Our zero-incoherence capacity is the corresponding exact-consistency threshold for modifiable encodings.

Definition II.28 (Zero-Incoherence Capacity)**.** The *zero-incoherence capacity* of an encoding system is:

$$\begin{aligned} C_0 = \sup\{R : &\text{ independent rate } R \\ &\Rightarrow \text{incoherence probability} = 0\} \end{aligned}$$

where "incoherence probability = 0" means no incoherent state is reachable under any modification sequence.

Theorem II.29 (Zero-Incoherence Threshold)**.** *The zero-incoherence capacity of any encoding system under independent modification is exactly 1:*

$$C_0 = 1$$

(L: CAP1, CAP2, CAP3)

We state the result in achievability/converse form because later sections attach the nontrivial converse and complexity consequences to the same threshold.

Theorem II.30 (Achievability)**.** *Independent rate 1 achieves zero incoherence. Therefore $C_0 \geq 1$.*

Proof. Let DOF$(C, F) = 1$. By Definition II.20, exactly one location L_s is independent; all other locations $\{L_i\}$ are derived from L_s.

By Definition II.19, derived locations satisfy: $\text{update}(L_s) \Rightarrow \text{automatically\_updated}(L_i)$. Therefore, for any reachable state:

$$\forall i : \text{value}(L_i) = f_i(\text{value}(L_s))$$

where f_i is the derivation function. All values are determined by L_s. Disagreement requires two locations with different determining sources, but only L_s exists. Therefore, no incoherent state is reachable. ■

Theorem II.31 (Converse)**.** *Any independent rate greater than* 1 *fails to achieve zero incoherence. Therefore $C_0 < R$ for all $R > 1$.*

Proof. Let $\text{DOF}(C, F) = k > 1$. By Definition II.18, there exist locations L_1, L_2 that can be modified independently.

Construction of incoherent state: Consider the modification sequence:

1) δ_1: Set $L_1 = v_1$ for some $v_1 \in \mathcal{V}_F$
2) δ_2: Set $L_2 = v_2$ for some $v_2 \neq v_1$

Both modifications are valid (each location accepts values from $\mathcal{V}_F$). By independence, δ_2 does not affect L_1. The resulting state has:

$$\text{value}(L_1) = v_1 \neq v_2 = \text{value}(L_2)$$

By Definition II.3, this state is incoherent. Since it is reachable, zero incoherence is not achieved.

The converse is intentionally combinatorial: once two independently writable locations exist, one can construct a reachable disagreement state. Later sections refine the same obstruction using finite zero-error side-information counting arguments. ■

Proof of Theorem II.29. By Theorem II.30, $C_0 \geq 1$.

By Theorem II.31, $C_0 < R$ for all $R > 1$.

Therefore $C_0 = 1$ exactly. The bound is tight: achieved at $\text{DOF} = 1$, not achieved at any $\text{DOF} > 1$. ■

Relation to Shannon capacity. The analogy is structural rather than literal: Shannon studies zero-error communication over confusable channel outputs, whereas here the object is exact consistency under admissible edits. The role of "rate" is played by the number of independently writable locations.

Corollary II.32 (Capacity-Achieving Rate is Unique)**.** *Independent rate* 1 *is the unique capacity-achieving encoding rate. No alternative achieves zero incoherence at higher rate.*

Corollary II.33 (Redundancy Above Capacity)**.** *Any encoding with $\rho > 0$ (redundancy) operates above capacity and cannot guarantee coherence.*

J. Side Information for Resolution

When an encoding system is incoherent, exact resolution requires external side information.

Theorem II.34 (Side Information Requirement)**.** *Given a state whose ambiguity class has size k, resolving to the correct value requires at least $\log_2 k$ bits of side information.*

(L: CIA1)

Proof. The ambiguity class presents k internally supported alternatives. By Theorem II.5, no internal rule distinguishes them.

Let $S \in \{1, \ldots, k\}$ denote the index of the correct source. The decoder must determine S to resolve correctly. Since S is uniformly distributed over k alternatives (by symmetry of the incoherent state):

$$H(S) = \log_2 k$$

Any resolution procedure is a function $R : \text{State} \to \mathcal{V}_F$. Without side information Y:

$$H(S|R(\text{State})) = H(S) = \log_2 k$$

because R can be computed from the state, which contains no information about which source is correct.

With side information Y that identifies the correct source:

$$H(S|Y) = 0 \Rightarrow I(S;Y) = H(S) = \log_2 k$$

Therefore, side information must expose at least $\log_2 k$ bits' worth of distinguishable tags. ■

Corollary II.35 (Unit Rate Requires Zero Side Information)**.** *At independent rate* 1*, resolution requires 0 bits of side information.*

Proof. With $k = 1$, there is one independent location. $H(S) = \log_2 1 = 0$. No uncertainty exists about the authoritative source. ■

Corollary II.36 (Side Information Scales with Redundancy)**.** *The required side information equals $\log_2(1 + \rho)$ where ρ is encoding redundancy:*

$$\textit{Side information required} = \log_2(\textit{DOF}) = \log_2(1 + \rho)$$

III. Derivation as the Capacity-Achieving Structure

Section II identifies the sharp threshold: exact consistency is guaranteed only at independent rate 1. Here the corresponding structure is *derivation*: one location carries the independent value, while all other representations are deterministic views of that source.

A. Rate 1 as the Exact-Consistency Regime

Definition III.1 (Optimal Encoding (Unit Independent Rate))**.** An encoding system C achieves the *optimal encoding rate* for fact F iff

$$\text{DOF}(C, F) = 1.$$

Rate 1 is the unique independent rate compatible with zero incoherence in the model. A single authoritative source with deterministic views is the corresponding structural form.

Corollary III.2 (Unit Rate Guarantees Determinacy)**.** *If $\textit{DOF}(C, F) = 1$, then the value of F is determinate in every reachable state.*

Corollary III.3 (Unit Rate Achieves $O(1)$ Update)**.** *If $\textit{DOF}(C, F) = 1$, then coherence restoration requires $O(1)$ manual updates.*

Corollary III.4 (Uniqueness of the Optimal Rate)**.** *$\textit{DOF} = 1$ is the unique rate guaranteeing exact consistency.*

Corollary III.5 (Incoherence Under Redundancy)**.** *If $DOF(C, F) > 1$, then incoherent states are reachable.*

Proof. Immediate from Theorem II.8. ■

B. Counting Converse Interpretation

The threshold and the converse fit together in a compact way.

- When ambiguity classes are trivial, no auxiliary information is needed and rate 1 suffices.
- When a nontrivial ambiguity class survives the observation transcript, Theorem VI.4 shows that exact recovery requires a sufficiently large side-information alphabet.
- If that side information is unavailable but exact correctness is still demanded, the architecture must rely on additional independently accessible support, which moves the system above the rate-1 regime and leads to the update-cost lower bound of Section VI.

Derivation is therefore the capacity-achieving dependence structure: it removes independent rate without removing observable views.

C. Rate–Complexity Tradeoff

The benefit of derivation is operational. An encoding may expose many visible copies of a fact, but only the independent copies govern manual synchronization cost.

- **Independent rate 1:** one manual update changes the authoritative source; all derived views follow automatically.
- **Independent rate above 1:** each independent location must be synchronized manually to restore coherence.

This is the architectural meaning of the threshold: independent rate above 1 creates both ambiguity and manual synchronization burden.

D. Derivation: The Relevant Dependence Structure

Definition III.6 (Derivation)**.** Location L_{derived} is *derived from* L_{source} for fact F iff an update to L_{source} determines the value of L_{derived} without an additional manual edit.

Derivation may occur at creation time, build time, commit time, or query time depending on the host system. The abstract point is unchanged: a derived location does not contribute an additional independently writable value for the same fact.

Theorem III.7 (Derivation Preserves Coherence)**.** *If $L_{derived}$ is derived from L_{source}, then $L_{derived}$ cannot diverge from L_{source} and does not contribute to DOF.*

(L: DER2)

Proof. By Definition III.6, the value at L_{derived} is determined by the value at L_{source} under admissible updates. There is therefore no reachable state in which the two locations carry incompatible values for the same fact. Since the derived location has no independently writable contribution, it does not increase DOF. ■

Corollary III.8 (Derivation Achieves the Threshold)**.** *If all encodings of F except one are derived from that one, then $DOF(C, F) = 1$ and exact consistency is guaranteed.*

Proof. Only one source remains independent; all other encodings are deterministic views. Apply Theorem II.7. ■

The same dependence structure can be realized in many host systems, but the theorem is purely about deterministic dependence in the abstract encoding model.

IV. Realizability Requirements for the Capacity-Achieving Regime

The threshold theorem identifies the unique independent rate compatible with exact consistency. A separate question is whether a concrete host system can realize that rate. This section isolates the two structural properties needed for that purpose.

A host system realizes the rate-1 regime when one location is authoritative and every secondary representation is maintained as a deterministic view. Two capabilities are required:

1) **Causal update propagation**: updates to the authoritative source must automatically propagate to derived locations.
2) **Provenance observability**: the system must expose enough structural information to verify which locations are authoritative and which are derived.

A. Confusability and Side Information

To connect realizability to information available at verification time, fix a fact F and consider the alternatives that remain compatible with the available observations. These alternatives form the same kind of zero-error obstruction studied earlier.

Lemma IV.1 (Confusability Clique Bound)**.** *If the confusability graph for F contains a clique of size k, then any zero-error side-information scheme distinguishing those k alternatives requires at least k distinct tags, equivalently at least $\log_2 k$ bits of side information.*

(L: OBS5)

Proof. Within a k-clique, every pair of alternatives is confusable under the base observation. A zero-error tag assignment must therefore separate all k states. Distinct states cannot share a tag, so at least k tags are required. ■

If a host system does not expose enough structural information to separate the remaining alternatives, exact verification of the rate-1 regime is impossible.

B. The Structural Timing Constraint

Definition IV.2 (Structural Fact)**.** A fact F is *structural* if every location encoding F is fixed when the underlying object, declaration, or artifact is created. After that moment, the encoding can be read and compared, but not retroactively recreated without a new edit event.

Structural facts are the natural exact-consistency setting because they expose a clear creation moment. Examples include schema declarations, registry membership rules, dependency metadata, and interface-level constraints.

Theorem IV.3 (Timing Constraint for Structural Derivation)**.** *For structural facts, derivation must occur no later than the moment at which the relevant encoding locations become fixed.*

Proof. Let t_{fix} denote the time at which the structural encoding becomes fixed. A derivation performed strictly before t_{fix} cannot depend on the completed source value; a derivation performed strictly after t_{fix} cannot alter the already fixed structural representation. Therefore structural derivation must occur at t_{fix} or as part of the same creation/update event. ■

C. Requirement 1: Causal Update Propagation

Definition IV.4 (Causal Update Propagation)**.** An encoding system has *causal update propagation* if every update to a source location automatically updates each location derived from that source, without requiring a separate manual synchronization step.

This is the host-system manifestation of deterministic dependence. Without it, a temporal gap appears between source modification and derived-view repair.

Theorem IV.5 (Causal Propagation is Necessary for Unit Rate)**.** *Achieving independent rate* 1 *for structural facts requires causal update propagation.*

(L: REQ2)

Proof. By Theorem IV.3, structural derivation must occur at the moment the source-side structural encoding is fixed. If causal propagation is absent, then some derived location remains stale until a later manual action. During that interval the source and derived locations disagree, so the architecture does not realize exact consistency. Hence a verifiable rate-1 realization for structural facts requires causal propagation. ■

D. Requirement 2: Provenance Observability

Definition IV.6 (Provenance Observability)**.** An encoding system has *provenance observability* if it supports queries that reveal which locations encode a fact, which of those locations are authoritative, and which are derived.

Provenance observability is the structural side information needed to certify that the system is genuinely in the single-source regime rather than merely appearing coherent on a small sample of states.

Theorem IV.7 (Provenance Observability is Necessary for Verifiable Unit Rate)**.** *Verifying that independent rate* 1 *holds requires provenance observability.*

(L: REQ3)

Proof. To verify independent rate 1, one must enumerate the locations encoding the fact, identify the authoritative source, and confirm that every remaining location is derived from it. Without provenance queries, these checks cannot be completed from inside the host system. The architecture may be coherent on observed states, but the single-source claim is not verifiable. ■

E. Independence of the Two Requirements

The two requirements are logically separate.

Theorem IV.8 (Requirements are Independent)**.**
1) *An encoding system can have causal propagation without provenance observability.*
2) *An encoding system can have provenance observability without causal propagation.*

(L: IND1, IND2)

Proof. For (1), consider a system that automatically refreshes all derived views after each source update but exposes no query interface for its derivation graph. Coherence may hold operationally, but the single-source claim is not inspectable. For (2), consider a system that records complete provenance metadata yet requires users to trigger propagation manually after each change. The derivation structure is visible, but stale views remain reachable. Hence neither property implies the other. ■

F. The Realizability Theorem

Theorem IV.9 (Necessary and Sufficient Realizability Conditions)**.** *An encoding system can achieve verifiable independent rate* 1 *for structural facts if and only if it provides both causal update propagation and provenance observability.*

(L: SOT1)

Proof. Necessity follows from Theorems IV.5 and IV.7. For sufficiency, assume both properties hold. Causal propagation ensures that every derived location is updated as part of the same structural event as the source; provenance observability then certifies that all secondary encodings are in fact derived from that source. The architecture therefore realizes a verifiable single-source encoding, i.e., independent rate 1. ■

The application section specializes the theorem to computational examples.

V. Application: Computational Realizations

This section records representative computational instantiations of Theorem IV.9.

Corollary V.1 (Computational-Realization Criterion)**.** *A computational platform realizes verifiable independent rate* 1 *for structural facts if and only if it combines automatic propagation of derived representations with queryable provenance for those representations.*

(L: LNG1)

The corollary is a direct restatement of Theorem IV.9. What changes across applications is the implementation of the two required capabilities.

Domain	Authoritative source and derived views	How the theorem specializes
Language-level artifacts	one declarative definition plus maintained registries, interface views, or metadata projections	definition-time or build-time hooks supply propagation; reflection or metadata APIs supply provenance
Databases	one base relation or schema declaration plus maintained views, indexes, or summaries	transactional maintenance supplies propagation; system catalogs expose provenance
Configuration and deployment systems	one source declaration plus dependency outputs or rendered manifests	dependency-driven recomputation supplies propagation; resource graphs expose provenance

Platforms that rely solely on external batch synchronization or opaque code generation do not automatically satisfy the corollary: if propagation can be bypassed or provenance is not available to the host system, exact verification of the single-source regime remains unresolved. Supplement A contains additional case-study material.

VI. Rate-Complexity Bounds

The threshold $C_0 = 1$ has an operational consequence: independent rate governs manual synchronization cost. This section states that consequence in the same finite, deterministic model used for the zero-error theorems.

A. Cost Model

Definition VI.1 (Modification Cost Model)**.** Let δ_F be a modification to fact F in encoding system C. The *effective modification complexity* $M_{\text{effective}}(C, \delta_F)$ is the number of locations that must be edited manually to preserve correctness after applying δ_F.

Only manual edits are counted. Derived locations updated automatically by the host system contribute zero effective cost.

B. Upper Bound at Unit Independent Rate

Theorem VI.2 (Rate-1 Upper Bound)**.** *If DOF$(C, F) = 1$, then*

$$M_{\text{effective}}(C, \delta_F) = O(1).$$

(L: BND1)

Proof. When $\text{DOF}(C, F) = 1$, there is exactly one independently writable source location for F. Updating that location is one manual edit; every other representation of F is derived and is updated automatically. The number of manual edits therefore stays bounded by a constant independent of the number of visible encodings. ■

C. Lower Bound Above the Threshold

Theorem VI.3 (Above-Threshold Lower Bound)**.** *If fact F is encoded at n independent locations, then*

$$M_{\text{effective}}(C, \delta_F) = \Omega(n).$$

(L: BND2)

Proof. Each independent location can retain an outdated value unless it is edited directly. To restore exact consistency after changing F, each of the n independent locations must therefore be synchronized manually. Hence the effective modification complexity grows at least linearly in n. ■

D. Finite Counting Converse

The update-cost lower bound is the operational face of the same information obstruction studied earlier: insufficient side information prevents exact recovery of the correct latent value.

Theorem VI.4 (Finite Counting Converse)**.** *Let F range over K possible latent values. Suppose exact recovery is attempted from a fixed observation transcript together with an L-bit side-information tag. Then*

$$K \le 2^L.$$

Equivalently, exact zero-error recovery of K ambiguous states requires at least $\log_2 K$ bits of side information.

(L: CIA3)

Proof. An L-bit tag yields at most 2^L distinct labels. If the base observation transcript is fixed across the ambiguous states, exact recovery requires the tag assignment to distinguish all K possibilities. Distinct states therefore need distinct tags, which is possible only when $K \le 2^L$. ■

Lemma VI.5 (Information-Constrained DOF Lower Bound)**.** *If the available side-information mechanism exposes at most 2^L distinguishable tags while the latent ambiguity class has size $K > 2^L$, then no zero-error resolver can identify the true state from those tags alone. Any architecture that still guarantees exact correctness must therefore rely on additional independently accessible discriminating support, moving the system above the single-source regime.*

(L: CIA4)

Proof. Theorem VI.4 rules out exact recovery from the available tag budget. If exact correctness is nevertheless required, the architecture must supplement the insufficient side-information channel with additional independently accessible information. In the present model, that means leaving the rate-1 regime and paying the synchronization cost of Theorem VI.3. ■

E. Worked Example

Consider a fact with $K = 4$ admissible values while the available side-information channel supplies only one bit. Since $2^1 < 4$, Theorem VI.4 rules out zero-error recovery. The architecture must either enlarge the side-information alphabet or expose additional independent support. The theorem does not depend on software; the same logic applies to any host system whose observation transcript leaves four mutually compatible alternatives.

F. Unified Converse Viewpoint

The counting theorem is the basic form of the converse, but it is not the only one. In the deterministic finite model, the same budget obstruction can be restated in several classical languages.

Theorem VI.6 (Unified finite converse, summary form)**.** *Let X be the latent source, let Y denote the deterministic observation/side-information transcript, and let $\hat{X}$ be any deterministic decoder output. If Y does not identify the true state exactly, then the same finite budget simultaneously:*

1) *lower-bounds zero-error ambiguity resolution and confusability-clique separation,*
2) *upper-bounds the conditional entropy $H(X \mid Y)$,*
3) *upper-bounds the decoder-output entropy $H(\hat{X})$,*
4) *and appears as an output-entropy or KL gap relative to the ambient finite alphabets.*

These are different normal forms of one deterministic finite converse family rather than unrelated lemmas.

The zero-incoherence threshold identifies the unique exact-consistency regime; the converse family explains how departures from that regime reappear as finite information-budget obstructions.

G. The Unbounded Gap

Theorem VI.7 (Unbounded Gap)**.** *The ratio between the modification cost of an architecture with n independent encodings and the modification cost of a rate-1 architecture diverges:*

$$\lim_{n\to\infty} \frac{M_{\text{DOF}>1}(n)}{M_{\text{DOF}=1}} = \infty.$$

(L: BND3)

Proof. By Theorem VI.2, a rate-1 architecture has constant effective modification complexity. By Theorem VI.3, an architecture with n independent encodings incurs linear cost in n. The ratio therefore grows without bound. ■

The asymptotic statement has a simple practical reading: once a fact is duplicated across many independently writable locations, the cost of preserving exact consistency scales with the number of independent copies, not merely with the visibility of the fact.

VII. Application: Supplementary Case Study Pointer

Supplement A contains case-study details, measurement methodology, and before/after traces. The examples realize independent rate 1 by combining one authoritative representation with automatically maintained derived views, and they exhibit the $O(1)$ versus $\Omega(n)$ rate-complexity gap of Section VI.

VIII. Related Work

A. Zero-Error and Side-Information Lineage

Shannon's zero-error framework and its graph-theoretic refinements by Körner and Lovász provide the closest conceptual starting point [1]–[3]. Witsenhausen's zero-error side-information problem is also directly relevant because it studies exact recovery under side information through graph coloring and label budgets [4]. Here the confusable objects are reachable states of a modifiable multi-location encoding rather than channel inputs. The threshold $C_0 = 1$ is therefore a storage/update analogue of a zero-error feasibility boundary.

Slepian–Wolf coding and classical entropy converses supply the second line of influence [5], [6], [9], [10]. Coding-for-computing and functional-compression results are also close neighbors because they study deterministic decoder targets under side information and graph constraints [7], [8]. In our setting the side information is deterministic rather than stochastic, but it plays the same operational role: exact recovery becomes impossible when the available observation/tag alphabet is too small to separate the remaining alternatives.

B. Consistency-Constrained Storage and Interaction

Consistency-constrained storage problems, including multi-version coding and related distributed-storage converses, show that correctness requirements impose unavoidable information costs [12]. Here the object under study is a modifiable encoding architecture, and the main question is the exact-consistency cost of multiple independently writable representations of one latent fact.

C. Computational Realizations

Reflection, metadata systems, and maintenance mechanisms in host platforms provide examples of how the realizability conditions can be instantiated in practice [13], [14].

D. Formal Verification

The Lean 4 formalization places the work in the tradition of mechanized mathematics and verified theory development [11], [15].

IX. Conclusion

Exact consistency in modifiable multi-location encodings yields a deterministic finite converse problem. Once the observation transcript leaves a nontrivial ambiguity class, exact recovery requires side information of sufficient cardinality, and the same finite budget reappears in confusability, conditional-entropy, decoder-output, and entropy-gap formulations.

Exact k-way ambiguity requires at least $\log_2 k$ bits of side information. The zero-incoherence capacity of the model is exactly one independent source: one authoritative location with deterministic views achieves exact consistency, while any higher independent rate makes ambiguity reachable. This threshold has an operational consequence: manual update cost is $O(1)$ at rate 1 and $\Omega(n)$ above it, with an unbounded asymptotic gap. The realizability theorem identifies causal propagation and provenance observability as the conditions for attaining the rate-1 regime in host systems.

These results describe one obstruction: insufficient information to distinguish competing latent states exactly.

Limitations and future work. The present model is finite and exact. Natural extensions include probabilistic coherence, networked encodings with communication constraints, and partial-correlation regimes that move closer to classical stochastic side-information models.

A. Artifacts

The Lean 4 formalization is included as supplementary material. Sections V and VII record computational instantiations and case-study material.

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Appendix

Mechanized Verification (Lean 4)

The main theorem chains are machine-checked in Lean 4. The artifact covers four groups of claims used in the paper:

- the foundational threshold chain, including coherence, independent rate, the side-information lower bound, and the finite counting converse;
- the realizability chain, including derivation, the timing constraint, causal propagation, provenance observability, and the realizability iff theorem;
- the rate-complexity chain, including the $O(1)$ upper bound, the $\Omega(n)$ lower bound, and the unbounded-gap consequence;
- the finite entropy layer supporting the conditional-entropy, decoder-output, and KL-gap reformulations.

The supplementary artifact contains the theorem-to-declaration coverage matrix, proof inventory, and build instructions.

Lean Handle Index.

BND1 ssot_upper_bound
BND2 non_ssot_lower_bound
BND3 ssot_advantage_unbounded
CAP1 ClaimClosure.cap_encoding_zero_error
CAP2 ssot_guarantees_coherence
CAP3 non_ssot_permits_incoherence
CIA1 ClaimClosure.side_information_requirement
CIA3 ClaimClosure.finite_counting_converse
CIA4 ClaimClosure.info_dof_counting_contradiction
COH1 dof_one_implies_coherent
COH2 dof_gt_one_incoherence_possible
DER2 ClaimClosure.derivation_preserves_coherence_core
IND1 both_requirements_independent
IND2 both_requirements_independent'
LNG1 ClaimClosure.language_realizability_criterion
OBS5 ObserverModel.clique_card_le_tag_alphabet
REQ2 definition_hooks_necessary
REQ3 introspection_necessary_for_verification
SOT1 ssot_iff